5 Week 5: Business Correspondence

5.1 Academic Integrity, Cyberethics, and Responsible AI in Computing

The integrity of an IT research project or graduation thesis relies on the ethical design, deployment, and documentation of the computing solution. In computing disciplines, **academic integrity and intellectual property (IP)** extend beyond text to include algorithms, source code, data structures, and trained model weights.

- **The Digital Workspace Dilemma:** Modern IT research relies heavily on third-party code libraries, public repositories, and generative AI models. Without strict compliance frameworks, a software artifact shifts from a piece of peer-reviewed engineering research into an illegal or plagiarized product.
- **Applying the Swales & Feak Paradigm:** Following Swales and Feak (2012), academic transparency requires providing exact attribution for the baseline components used to construct a system. In IT, this means explicitly delineating between what you coded and what pre-existed in the engineering domain.

5.2 Plagiarism Typologies in IT Engineering

Plagiarism in computing research encompasses both textual narrative fraud and programmatic theft. Students must be trained to identify four core technical typologies:

1. **Direct Plagiarism (Text & Source Code Verbatim Copying)**
   - **Definition:** Copying text, mathematical models, or source code blocks exactly from a source without quotation marks or citation.
   - **Example:** A researcher copies an entire data scrubbing function from a GitHub repository or lifts a paragraph explaining neural layers directly from a paper, pasting it into their thesis without an inline citation block.
2. **Mosaic Plagiarism (Patchwork Engineering)**
   - **Definition:** Patching together phrases, code snippets, or algorithm steps from multiple online sources, changing a few variable names or words, while keeping the original logical structure without attribution.
   - **Example:** Latching onto three different answers on StackOverflow, blending their variable names (int cloud_temp changed to int t) , and merging them into one function without comments acknowledging the source contributors.
3. **Self-Plagiarism (Dual Submission)**
   - **Definition:** Reusing one's own previous code or text from a prior course assignment or publication without explicit disclosure or authorization.
   - **Example:** Re-submitting a mobile UI database module developed during a 3rd-year Mobile Applications course as an original system component for the 4th-year graduation thesis without documenting it as a baseline asset.
4. **Accidental Plagiarism (Careless Documentation)**
   - **Definition:** Neglecting to cite an asset or library due to systemic sloppiness, losing track of open tabs, or failing to understand citation syntax guidelines.
   - **Example:** Utilizing a public API wrapper inside an application backend, intending to add the citation later, but ultimately forgetting to declare it in the references before submission.

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5.3 Technical Specifications of Standard Citation Styles: IEEE vs. APA

Technical documentation relies on absolute accuracy when attributing sources. Students must master the internal structures of both **IEEE** (technical standard) and **APA** (socio-technical standard) formats.

1. **IEEE Citation Style (Numeric Engine)**
    - **In-Text Bracket Engine:** Uses sequential numbers enclosed in square brackets [1] matching the precise order of appearance in the text.
    - Reference Entry Mechanics: Entries are organized numerically, not alphabetically. Author initials precede surnames.
    - **IEEE Formats for IT:**
        + **Journal Article:** J. K. Author, "Name of paper," Abbrev. Title of Journal, vol. x, no. x, pp. xxx–xxx, Mon. Year.
        + **Conference Proceeding:** A. B. Researcher, "Title of software paper," in Proc. Name of Conf., Year, pp. xx–xx.
        + **Software/Code Repository:** Developer Name or Company, "Title of software package/repository," Platform Name, Year. URL.
2. **APA Citation Style (Author-Date Engine)**
    - **In-Text Parenthetical Engine:** Uses the author's last name and the year of publication inside parentheses (Smith, 2026).
    - **Reference Entry Mechanics:** Entries are ordered alphabetically by the author's last name. Author surnames precede initials.
    - **APA Formats for IT:**
        + **Journal Article:** Author, A. B. (Year). Title of paper. Title of Journal, x(x), xxx–xxx.
        + **Conference Proceeding:** Researcher, A. B. (Year). Title of software paper. Proceedings of the Conf Name, xxx–xxx.
        + **Software/Code Repository:** Developer Name or Company. (Year). Title of software package/repository (Version x.x) [Computer software]. Platform Name. URL.

Comparative Syntax Matrix: IEEE vs. APA

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Scenario:  Citing an AI research paper written by Dalya Raad AlAni in 2026

   IEEE Style:
In-Text:  "The predictive neural model follows the optimization path defined in
[1]."
Reference List:
[1] D. R. AlAni, "Predictive neural architectures for edge computing," IEEE Trans.
Comput., vol.  12, no.  3, pp.  45-52, Mar.  2026.

   APA Style:
In-Text:  "The predictive neural model follows the optimization path defined in
(AlAni, 2026)."
Reference List:
AlAni, D. R. (2026).  Predictive neural architectures for edge computing.  IEEE
Transactions on Computers, 12(3), 45-52.
```

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5.4 Data Privacy and Cybersecurity Compliance Frameworks: GDPR & ISO

When an IT system handles personal identifiers, metrics, or telemetry, it operates within absolute legal and security bounds.

1. **General Data Protection Regulation (GDPR)**

   The **GDPR** is an active, binding European regulation that applies globally to any computing system processing the personal data of individuals residing in the EU. Students must integrate two core principles into their technical architecture blueprints:

   - **Data Minimization:** Systems must restrict data collection to the absolute minimum necessary for function. If a system needs a user account, it must not collect telemetry variables like GPS coordinates unless directly required.
   - **Privacy by Design:** Security and anonymity features must be coded directly into the baseline foundation of the application, rather than tacked on as an afterthought. For example, automated salting and hashing mechanisms for user passwords inside the database layer layout.

2. **ISO/IEC 27001 Standard**

   ISO/IEC 27001 is the leading international standard governing an Information Security Management System (ISMS). In an IT graduation project, it provides the structural checklist for managing risks around data assets:

   - **Access Control Layouts:** Documenting user role authentications (e.g., RBAC blueprints) within the system design chapter to enforce the principle of least privilege.
   - **Cryptographic Controls:** Explicitly stating the encryption protocols (e.g., AES-256 for data-at-rest, TLS 1.3 for data-in-transit) within the technical environment framework.

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5.5 Sub-Themes & Application: Academic Honesty & Code Attribution

Use this matrix to clarify how abstract digital regulations translate into actual engineering standards:

Concept Framework	Academic/IT Infraction	Correct Professional Documentation Practice
Academic Honesty Paradigm	**Data Fabrication:** Inventing performance metrics (e.g., latency, accuracy) because the system failed under load testing.	Document the actual system failures honestly in the evaluation chapter; analyze the constraints as a research finding.
Open-Source Compliance	**License Violation:** Bundling a GPLv3 network wrapper inside a commercialized or closed-source proprietary app.	Disclose the repository dependencies, append the original copyright text notices, and match the target licensing model.
Code Attribution Mechanics	**Shadow Code Adoption:** Pasting a complex image-processing function directly from a repository without comments.	Place a structured attribution block directly inside the source code comments containing the author, year, URL, and DOI.
GDPR Compliance Framework	**Illegal Logging:** Storing unencrypted user email addresses and IP addresses in plaintext system error logs.	Embed a data-scrubbing regex pipeline that strips or hashes identifiable tracking parameters before data persistence.
ISO/IEC 27001 Controls	**Hardcoded Credentials:** Leaving database passwords and API master keys written directly inside source scripts.	Utilize environmental configuration variables (*.env*) managed via an external secure key management service.

Table 1: Expanded Concept Application Table

5.6 Complete IT Example: Automated Cloud Infrastructure Ledger

- **Project:** "Designing a Secure Microservice Mesh with Automated Compliance Logging"
- **Academic Honesty Framework:** The researcher uses a pre-trained encryption library to secure the service mesh communication. Rather than claiming the encryption script as an original creation, the researcher explicitly documents the library's developer using an IEEE citation bracket within the system architecture description.
- **Open-Source Compliance Model:** The core logging architecture utilizes an Apache 2.0 licensed software component. The researcher maintains the original license file within their public GitHub repository directory, ensuring absolute transparency regarding third-party intellectual property.
- Data Privacy Integration: Because the ledger records system logs that could accidentally capture client IP addresses, the system design implements a GDPR Privacy-by-Design rule. A data scrubbing function automatically hashes personal identifiers before they write to the persistent storage layer.Because the ledger records system logs that could accidentally capture client IP addresses, the system design implements a GDPR Privacy-by-Design rule. A data scrubbing function automatically hashes personal identifiers before they write to the persistent storage layer.
- **Generative AI Disclosure:** The researcher utilizes an AI assistant to generate the boilerplate Docker deployment configuration scripts. In the appendix of the thesis document, the researcher attaches a standard AI Usage Declaration Form, detailing the prompt parameters used to generate the infrastructure-as-code files.

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5.7 The Grammar of Compliance: Present Tense Protocol

IWhen writing about active cybersecurity laws, ethical frameworks, and institutional rules, researchers must utilize the **Present Simple Tense**. Because these regulations are actively binding and operational at the time of reading, past tense usage implies the law is dead or no longer enforced.

Target Subject	Tense Grammar	Active Formula	Student-Facing Example
Active Cybersecurity Laws	Present Simple	Subject + Verb 1 (s/es)	"**The GDPR mandates** that de ers implement strict data minimi workflows."
Ethical Compliance Frameworks	Present Simple	Subject + Verb 1 (s/es)	"**The IEEE Code of Ethics re** engineers to disclose conflicts of est."
Institutional System Rules	Present Simple	Subject + Verb 1 (s/es)	"Our **university policy prohibi** uncredited submission of AI-gen code."
Completed Code Attribution	Past Simple	Subject + Verb 2 (or Passive)	Contrast: "The third-party repo **was accessed** on October 10, 202

Table 2: Tense Protocol for Digital Compliance

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References

Swales, J. M. and Feak, C. B. (2012). *Academic Writing for Graduate Students: Essential Tasks and Skills*. University of Michigan Press, 3rd edition.

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